

Week 3 Internship Task

Clustering Models & Association Rule Mining

Objective of the Task

This week's task is divided into two parts. In the first part, you will explore customer segmentation by applying multiple clustering algorithms to a real customer dataset, tuning their parameters, and evaluating cluster quality. In the second part, you will apply association rule mining to real transactional data, extracting frequent itemsets and rules using both the Apriori and FP-Growth algorithms, and comparing their performance. Together, these two parts represent the core workflow of unsupervised learning on real-world data: clustering to find hidden groups, and association rule mining to find hidden relationships.

Datasets

- Name (Part 1): Mall Customer Segmentation Dataset
 - Description: Customer records from a mall, containing Age, Annual Income, Spending Score, and Gender.
- Name (Part 2): Groceries Market Basket Dataset
 - Description: Real transactional data from a grocery store, containing itemized lists of products purchased per transaction.

Part 1 — Clustering Models

Requirements

1. Load the Mall Customer dataset and re-examine its structure (shape, dtypes, missing values, summary statistics).
2. Select relevant numerical features (Annual Income, Spending Score, Age) and scale them using StandardScaler.
3. Fit K-Means for a range of k values (1–10) and plot the Elbow Method curve to determine the optimal k.
4. Fit the final K-Means model with the chosen k, visualize the resulting clusters with a 2D scatter plot, and compute the Silhouette Score.
5. Plot a dendrogram using Ward linkage and fit Agglomerative (Hierarchical) Clustering with the number of clusters read off the dendrogram.
6. Fit DBSCAN, tuning eps and min_samples using a k-distance plot, and visualize the resulting clusters including noise points (label = -1).
7. Create a comparison table of K-Means, Hierarchical Clustering, and DBSCAN covering: assumptions, handling of outliers/noise, need to pre-specify the number of clusters, and scalability.
8. Write a 250-word Clustering Summary at the end covering: how many meaningful customer segments were found, which algorithm produced the most interpretable clusters, how each segment could be described in business terms (e.g. high income/low spending), and one hypothesis about what drives the spending behavior of the highest-value segment.

Part 2 — Association Rule Mining

Requirements

9. Load the Groceries dataset with all transactions and re-examine its structure.
10. Convert the transaction list into one-hot encoded format using `mlxtend.preprocessing.TransactionEncoder`.

11. Generate frequent itemsets using the Apriori algorithm (`mlxtend.frequent_patterns.apriori`) with a chosen `min_support` — document the value chosen and why.
12. Generate association rules from the Apriori itemsets and report support, confidence, and lift for each rule.
13. Generate frequent itemsets and rules using the FP-Growth algorithm (`mlxtend.frequent_patterns.fpgrowth`) with the same `min_support`, and confirm the results match Apriori.
14. Time both algorithms on the same dataset (e.g. using `time.time()` or `%timeit`) and report the runtimes in a table.
15. Vary `min_support` (e.g. 0.01, 0.02, 0.05) and `min_confidence` (e.g. 0.3, 0.5, 0.7) and record how the number of frequent itemsets and rules changes.
16. Create a bar chart comparing the top 10 rules by lift, with proper labels and title.
17. Write a Rule Mining Summary section in markdown explaining: which rules had the highest lift and what they mean in business terms, why FP-Growth is generally faster than Apriori (no candidate generation, compressed FP-tree vs. repeated database scans), and which algorithm you would carry forward for larger datasets.

Deliverables

- GitHub Repository Link – the repository must include:
 - Both notebooks: `week2_clustering.ipynb` and `week2_association_rules.ipynb`
 - `README.md` and `requirements.txt`
 - `data/` folder with the datasets